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Lecture 23

1. From a mathematical perspective, what is the primary difference between calculating electric fields for point charges versus continuous charge distributions?
 - a) Point charges require vector addition while continuous distributions need scalar addition
 - b) Point charges use summation while continuous distributions require integration
 - c) Point charges use multiplication while continuous distributions

use division

d) There is no mathematical difference

Answer: b) Point charges use summation while continuous distributions require integration

2. From a dimensional analysis viewpoint, which statement about charge density is correct?

a) Linear charge density (λ) has units of C/m

b) Surface charge density (σ) has units of C/m³

c) Volume charge density (ρ) has units of C/m

d) All charge densities have the same units

Answer: a) Linear charge density (λ) has units of C/m

3. From an experimental physicist's perspective, what happens to the electric field of a charged ring at very large distances?

a) It becomes zero

b) It becomes infinite

c) It behaves like a point charge

d) It remains unchanged

Answer: c) It behaves like a point charge

4. From a theoretical physics viewpoint, what is the fundamental principle behind Gauss's Law?

a) Conservation of energy

b) Conservation of charge

c) The relationship between electric flux and enclosed charge

d) Conservation of momentum

Answer: c) The relationship between electric flux and enclosed charge

5. From an engineering perspective, what is the key advantage of using Gauss's Law over Coulomb's Law?

- a) It works for any shape of closed surface
- b) It only works for spherical surfaces
- c) It requires less mathematical computation
- d) It gives more accurate results

Answer: a) It works for any shape of closed surface

6. From a field theory perspective, what is the significance of the electric flux through a closed surface?

- a) It represents the total work done by the field
- b) It measures the total charge enclosed within the surface
- c) It indicates the strength of magnetic field
- d) It shows the potential difference

Answer: b) It measures the total charge enclosed within the surface

7. From a problem-solving viewpoint, what happens to the electric field inside a hollow charged sphere?

- a) It equals zero
- b) It equals the surface charge density
- c) It varies linearly with radius
- d) It follows inverse square law

Answer: a) It equals zero

8. From an analytical perspective, what is the key characteristic of electric field lines at a distance r from a uniformly charged infinite wire?

- a) They are parallel to the wire
- b) They are perpendicular to the wire
- c) They form circular loops around the wire
- d) They are random in direction

Answer: b) They are perpendicular to the wire

9. From a mathematical physics perspective, what is the relationship between charge density and dimensionality of the charge distribution?

- a) Linear density (λ) = dq/ds**
- b) Surface density (σ) = dq/dV**
- c) Volume density (ρ) = dq/dA**
- d) All densities have same formula**

Answer: a) Linear density (λ) = dq/ds

10. From a conceptual viewpoint, why is Gauss's Law equivalent to Coulomb's Law for a point charge?

- a) Because both laws deal with electric fields**
- b) Because flux through a sphere gives the $1/r^2$ dependence**
- c) Because both laws involve constants**
- d) Because they use the same units**

Answer: b) Because flux through a sphere gives the $1/r^2$ dependence

11. From an applications perspective, what is the primary use of charge density concepts?

- a) To calculate fields from discrete charges only**
- b) To calculate fields from continuous charge distributions**
- c) To measure electric potential**
- d) To determine magnetic fields**

Answer: b) To calculate fields from continuous charge distributions

12. From a vector calculus perspective, what is the correct mathematical representation of electric flux?

- a) Simple product of E and A**
- b) Cross product of E and A**

- c) Surface integral of $E \cdot dA$
- d) Volume integral of $E \cdot dA$

Answer: c) Surface integral of $E \cdot dA$

- 13. From a symmetric distribution viewpoint, what happens to the electric field outside a uniformly charged sphere?**
- a) Behaves as if all charge is at center
 - b) Varies linearly with distance
 - c) Remains constant
 - d) Increases exponentially

Answer: a) Behaves as if all charge is at center

- 14. From a physical interpretation standpoint, what does charge density represent?**
- a) Total charge in a body
 - b) Charge distribution per unit space
 - c) Electric field strength
 - d) Potential difference

Answer: b) Charge distribution per unit space

- 15. From a computational perspective, why do we break continuous charge distributions into small elements?**
- a) To apply calculus principles
 - b) To treat each element as a point charge
 - c) To reduce computational errors
 - d) Both a and b

Answer: d) Both a and b

- 16. From a boundary conditions perspective, what happens to the electric field at the boundary of a charged conductor?**
- a) Field is perpendicular to the surface

- b) Field is parallel to the surface
- c) Field is zero at the boundary
- d) Field is maximum at the boundary

Answer: a) Field is perpendicular to the surface

17. From a superposition principle viewpoint, how do we handle multiple continuous charge distributions?

- a) Take the largest distribution only
- b) Average all distributions
- c) Add vector fields from each distribution
- d) Multiply the fields

Answer: c) Add vector fields from each distribution

18. From an integral calculus perspective, what makes calculating electric field of a continuous charge distribution challenging?

- a) Complex geometry
- b) Vector nature of the field
- c) Multiple integrations required
- d) All of the above

Answer: d) All of the above

19. From a symmetry analysis viewpoint, when is Gauss's Law most useful?

- a) For asymmetric charge distributions
- b) For highly symmetric charge distributions
- c) For point charges only
- d) For linear charges only

Answer: b) For highly symmetric charge distributions

20. From an experimental measurement perspective, what determines the direction of electric field lines?

- a) Direction of positive test charge movement
- b) Direction of negative test charge movement
- c) Direction of magnetic field
- d) Direction of current flow

Answer: a) Direction of positive test charge movement

21. From a dimensional analysis perspective, what is the relationship between E and σ (surface charge density)?

- a) $E = \sigma/2\epsilon_0$
- b) $E = \sigma/\epsilon_0$
- c) $E = 2\sigma/\epsilon_0$
- d) $E = \sigma\epsilon_0$

Answer: b) $E = \sigma/\epsilon_0$

22. From a field theory perspective, what is unique about electric field lines?

- a) They never cross each other
- b) They form closed loops
- c) They always point towards negative charges
- d) They are always parallel

Answer: a) They never cross each other

23. From an energy perspective, how is the work done related to moving a charge in an electric field?

- a) Always positive
- b) Always negative
- c) Path dependent
- d) Path independent

Answer: d) Path independent

24. From a vector calculus perspective, what is the significance of divergence of electric field?

- a) Represents charge density/ ϵ_0
- b) Always zero
- c) Represents current density
- d) Represents magnetic field

Answer: a) Represents charge density/ ϵ_0

25. From a practical application viewpoint, why is the field inside a hollow conductor zero?

- a) Due to charge neutrality
- b) Due to electrostatic shielding
- c) Due to magnetic effects
- d) Due to current flow

Answer: b) Due to electrostatic shielding

Lecture 24

1. From a fundamental concept perspective, what makes the electrostatic force conservative?

- a) Path independence of work done
- b) Dependence on distance only
- c) Both a and b
- d) Neither a nor b

Answer: c) Both a and b

2. From an energy conversion viewpoint, what happens when a positive charge moves in an electric field?

- a) Always gains potential energy
- b) Moves from high to low potential

- c) Moves from low to high potential
- d) No energy conversion occurs

Answer: b) Moves from high to low potential

3. From a mathematical perspective, how is electric potential related to electric field?

- a) $E = -\nabla V$
- b) $E = \nabla V$
- c) $E = V^2$
- d) $E = 1/V$

Answer: a) $E = -\nabla V$

4. From a dimensional analysis viewpoint, what is the SI unit of electric potential?

- a) Newton/Coulomb
- b) Joule/Coulomb
- c) Coulomb/meter
- d) Newton-meter

Answer: b) Joule/Coulomb

5. Regarding electron-volt as a unit, what does 1 eV physically represent?

- a) Energy of one electron
- b) Potential difference of 1 volt
- c) Energy gained by an electron through 1V potential difference
- d) Charge of an electron

Answer: c) Energy gained by an electron through 1V potential difference

6. From a superposition principle perspective, how do we calculate total potential due to multiple charges?

- a) Multiply individual potentials

- b) Take average of potentials
- c) Add algebraically
- d) Take the maximum value

Answer: c) Add algebraically

7. For an electric dipole, how does potential vary with distance r (far field)?

- a) $1/r$
- b) $1/r^2$
- c) $1/r^3$
- d) r

Answer: a) $1/r$

8. From a practical application viewpoint, why do negative charges move towards higher potential?

- a) Due to their negative charge
- b) Due to gravitational effects
- c) Due to magnetic effects
- d) Due to thermal effects

Answer: a) Due to their negative charge

9. From an energy conservation perspective, what happens to electric potential energy when charges are brought closer?

- a) Always increases
- b) Always decreases
- c) Increases for like charges, decreases for unlike charges
- d) Remains constant

Answer: c) Increases for like charges, decreases for unlike charges

10. In terms of field theory, what characterizes electric potential?

- a) It's a vector quantity

- b) It's a scalar quantity
- c) It's always positive
- d) It's always negative

Answer: b) It's a scalar quantity

- 11. From a charge distribution perspective, what is the relationship between potential and charge density?**
- a) Higher charge density always means higher potential
 - b) Higher positive charge density means higher potential
 - c) Higher negative charge density means higher potential
 - d) No relationship exists

Answer: b) Higher positive charge density means higher potential

- 12. From a force-energy relationship viewpoint, how is work done related to potential difference?**
- a) $W = q\Delta V$
 - b) $W = V/q$
 - c) $W = q/V$
 - d) $W = V^2$

Answer: a) $W = q\Delta V$

- 13. In terms of energy reference points, why is potential at infinity typically chosen as zero?**
- a) For mathematical convenience
 - b) Because no electric field exists there
 - c) Because it's a natural reference point for bounded systems
 - d) All of the above

Answer: d) All of the above

- 14. From a problem-solving perspective, what makes electric dipole potential calculations easier than field calculations?**

- a) Scalar nature of potential
- b) Vector nature of potential
- c) Lower order dependency on distance
- d) Higher order dependency on distance

Answer: a) Scalar nature of potential

15. For a continuous charge distribution, how is potential calculated?

- a) Through direct addition
- b) Through integration
- c) Through differentiation
- d) Through multiplication

Answer: b) Through integration

16. What physical principle explains why charges move spontaneously to minimize potential energy?

- a) Conservation of charge
- b) Conservation of energy
- c) Principle of least action
- d) Newton's first law

Answer: c) Principle of least action

17. From an experimental perspective, how can electric potential be measured?

- a) Using an ammeter
- b) Using a voltmeter
- c) Using a galvanometer
- d) Using a thermometer

Answer: b) Using a voltmeter

18. In terms of energy conversion, what happens when a charge moves between points of different potential?
- a) Kinetic energy changes to potential energy
 - b) Potential energy changes to kinetic energy
 - c) Both a and b are possible depending on direction
 - d) No energy conversion occurs

Answer: c) Both a and b are possible depending on direction

19. For a spherical conductor, how does potential vary with radius?
- a) Linearly inside the conductor
 - b) $1/r$ outside the conductor
 - c) Constant inside the conductor
 - d) Both b and c

Answer: d) Both b and c

20. From an applications perspective, why is potential difference more commonly used than absolute potential?
- a) Easier to measure
 - b) More practical significance
 - c) Independent of reference point
 - d) All of the above

Answer: d) All of the above

21. In terms of work done, what is special about a conservative force field?
- a) Work depends only on initial and final positions
 - b) Work depends on the path taken

- c) Work is always positive
- d) Work is always negative

Answer: a) Work depends only on initial and final positions

22. From a dimensional analysis perspective, what is the relationship between electric field and potential gradient?

- a) They have same units
- b) E has units of V/m
- c) V has units of E/m
- d) No relationship exists

Answer: b) E has units of V/m

23. For a system of point charges, what determines the sign of the total potential energy?

- a) Number of charges
- b) Distance between charges
- c) Types of charges and their separation
- d) Magnitude of charges only

Answer: c) Types of charges and their separation

24. When converting between electron volts and joules, which is correct?

- a) $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$
- b) $1 \text{ J} = 1.6 \times 10^{-19} \text{ eV}$
- c) $1 \text{ eV} = 1.6 \times 10^{19} \text{ J}$
- d) $1 \text{ J} = 1 \text{ eV}$

Answer: a) $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

25. In terms of equipotential surfaces, which statement is true?

- a) Electric field lines are parallel to them
- b) Electric field lines are perpendicular to them

- c) They never intersect with field lines
- d) Both b and c

Answer: d) Both b and c

26. From a field theory perspective, what happens to the potential between two oppositely charged parallel plates?

- a) Varies exponentially
- b) Varies linearly
- c) Remains constant
- d) Varies quadratically

Answer: b) Varies linearly

27. In terms of mathematical analysis, what is the significance of $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for electrostatic fields?

- a) Field is conservative
- b) Field is non-conservative
- c) Field is uniform
- d) Field is zero

Answer: a) Field is conservative

28. For an isolated conductor, how are charge and potential related?

- a) Charge is proportional to potential
- b) Charge is inversely proportional to potential
- c) No direct relationship exists
- d) Charge is proportional to square of potential

Answer: a) Charge is proportional to potential

29. From an energy storage perspective, where is the electrical potential energy stored?

- a) In the charges

- b) In the electric field
- c) In the conductor
- d) In the electrons

Answer: b) In the electric field

30. What fundamental principle explains why electric field lines must be perpendicular to conductor surfaces?

- a) Conservation of energy
- b) Conservation of charge
- c) No work done parallel to surface
- d) Gauss's law

Answer: c) No work done parallel to surface

31. For a dipole in a uniform electric field, what determines its potential energy?

- a) Only the dipole moment
- b) Only the field strength
- c) Product of dipole moment and field
- d) Product of dipole moment and field $\cos\theta$

Answer: d) Product of dipole moment and field $\cos\theta$

32. From a quantum mechanics perspective, what is the significance of electron-volt (eV)?

- a) Atomic energy level differences
- b) Nuclear binding energies
- c) Particle physics energies
- d) All of the above

Answer: d) All of the above

33. What happens to the potential energy when two like charges are separated to infinity?

- a) Increases to infinity
- b) Decreases to zero
- c) Remains constant
- d) Becomes negative

Answer: b) Decreases to zero

34. In terms of work done, why is potential a scalar quantity?

- a) Because work is a scalar
- b) Because charge is a scalar
- c) Because energy is a scalar
- d) All of the above

Answer: a) Because work is a scalar

35. When dealing with continuous charge distributions, what makes potential calculations often easier than field calculations?

- a) Scalar addition vs vector addition
- b) Lower order of complexity
- c) Simpler integration
- d) Both a and c

Answer: d) Both a and c

36. From an energy conservation viewpoint, what happens when a positive charge is released in an electric field?

- a) Moves to higher potential
- b) Moves to lower potential
- c) Stays stationary
- d) Moves perpendicular to field

Answer: b) Moves to lower potential

37. For a point charge, how does the ratio of electric field to potential vary with distance?

- a) $1/r$
- b) $1/r^2$
- c) r
- d) r^2

Answer: a) $1/r$

38. What determines the maximum potential difference that can exist between two points in air?

- a) Distance between points
- b) Humidity
- c) Temperature
- d) All of the above

Answer: d) All of the above

39. From a thermodynamic perspective, how does temperature affect the potential of a conductor?

- a) No effect
- b) Direct relationship
- c) Inverse relationship
- d) Logarithmic relationship

Answer: a) No effect

40. In terms of energy conversion in a capacitor, what form does electrical potential energy take?

- a) Chemical energy
- b) Field energy
- c) Thermal energy
- d) Kinetic energy

Answer: b) Field energy

BASIC CONCEPTS:

1. What fundamentally defines a capacitor?

- a) Single conductor storing charge**
- b) Two conductors with potential difference**
- c) Any device storing energy**
- d) Any two metal plates**

Answer: b) Two conductors with potential difference

2. The relationship $Q = CV$ indicates that:

- a) Charge is independent of voltage**
- b) Charge is inversely proportional to voltage**
- c) Charge is directly proportional to voltage**
- d) Charge varies exponentially with voltage**

Answer: c) Charge is directly proportional to voltage

MATHEMATICAL RELATIONSHIPS:

3. For parallel plate capacitor, capacitance C is given by:

- a) $C = \epsilon_0 A/d$**
- b) $C = \epsilon_0 d/A$**
- c) $C = \epsilon_0 Ad$**
- d) $C = \epsilon_0/Ad$**

Answer: a) $C = \epsilon_0 A/d$

4. In parallel combination of capacitors, total capacitance is:

- a) Less than individual capacitances**
- b) Sum of individual capacitances**
- c) Product of individual capacitances**
- d) Reciprocal of sum of reciprocals**

Answer: b) Sum of individual capacitances

PRACTICAL APPLICATIONS:

5. Capacitance can be increased by:

- a) Increasing plate separation
- b) Decreasing plate area
- c) Adding dielectric material
- d) Removing dielectric material

Answer: c) Adding dielectric material

ENERGY CONCEPTS:

6. The energy stored in a capacitor is:

- a) $CV^2/2$
- b) CV
- c) C^2V
- d) V^2/C

Answer: a) $CV^2/2$

7. The energy density in electric field is proportional to:

- a) E
- b) E^2
- c) $\sqrt{E}$
- d) $1/E$

Answer: b) E^2

DIELECTRIC EFFECTS:

8. When a dielectric is inserted between capacitor plates:

- a) Capacitance decreases
- b) Capacitance increases
- c) Capacitance remains same
- d) Capacitance becomes zero

Answer: b) Capacitance increases

9. The dielectric constant ϵ_r is:

- a) Always less than 1
- b) Always equal to 1
- c) Always greater than 1
- d) Can be any value

Answer: c) Always greater than 1

PROBLEM-SOLVING:

10. For series combination of capacitors:

- a) $1/C_{eq} = 1/C_1 + 1/C_2$
- b) $C_{eq} = C_1 + C_2$
- c) $C_{eq} = C_1 C_2$
- d) $1/C_{eq} = C_1 + C_2$

Answer: a) $1/C_{eq} = 1/C_1 + 1/C_2$

FIELD THEORY:

11. The electric field between capacitor plates is:

- a) Zero
- b) Uniform (ignoring edge effects)
- c) Varies linearly
- d) Varies exponentially

Answer: b) Uniform (ignoring edge effects)

12. When a capacitor is charged:

- a) Energy is stored in plates**
- b) Energy is stored in electric field**
- c) Energy is stored in battery**
- d) Energy disappears**

Answer: b) Energy is stored in electric field

PRACTICAL CONSIDERATIONS:

13. The effect of bringing capacitor plates closer is:

- a) Increased capacitance**
- b) Decreased capacitance**
- c) No change in capacitance**
- d) Zero capacitance**

Answer: a) Increased capacitance

14. In water molecules near a positive charge:

- a) Only positive ends align**
- b) Only negative ends align**
- c) Negative ends point toward charge**
- d) No alignment occurs**

Answer: c) Negative ends point toward charge

15. The dielectric constant of air is approximately:

- a) 0**
- b) 1**
- c) 1.0003**
- d) 8**

Answer: c) 1.0003

ADVANCED CONCEPTS:

16. For a given voltage, the charge stored in a capacitor with dielectric compared to without dielectric is:

- a) Same
- b) ϵ_r times more
- c) ϵ_r times less
- d) $1/\epsilon_r$ times

Answer: b) ϵ_r times more

17. The energy stored in capacitor is affected by dielectric because:

- a) Charge changes
- b) Voltage changes
- c) Both charge and voltage change
- d) Neither change

Answer: c) Both charge and voltage change

APPLICATIONS AND ANALYSIS:

18. When two identical capacitors are connected in series, the equivalent capacitance is:

- a) Double
- b) Half
- c) Same
- d) Four times

Answer: b) Half

19. The physical reason for increased capacitance with dielectric is:

- a) Increased conductivity**
- b) Decreased field strength**
- c) Molecular polarization**
- d) Increased plate area**

Answer: c) Molecular polarization

20. For a fixed charge, inserting a dielectric:

- a) Increases potential difference**
- b) Decreases potential difference**
- c) No effect on potential difference**
- d) Makes potential difference zero**

Answer: b) Decreases potential difference

THEORETICAL CONCEPTS:

21. The electric field reduction in a dielectric is due to:

- a) Reduction in charge**
- b) Internal polarization field**
- c) Increased conductivity**
- d) Decreased plate separation**

Answer: b) Internal polarization field

22. For a fixed voltage source, when a dielectric is inserted:

- a) Field strength increases**
- b) Field strength decreases**
- c) Charge increases**
- d) Both b and c**

Answer: d) Both b and c

PRACTICAL DESIGN:

23. Maximum voltage a capacitor can handle is limited by:

- a) Plate area
- b) Plate separation
- c) Dielectric breakdown
- d) Temperature only

Answer: c) Dielectric breakdown

24. In practical capacitors, edge effects are minimized by:

- a) Using smaller plates
- b) Using larger plates
- c) Guard rings
- d) Increasing separation

Answer: c) Guard rings

ENERGY ANALYSIS:

25. When a charged capacitor is disconnected from battery:

- a) Charge remains constant
- b) Voltage remains constant
- c) Energy remains constant
- d) All of above

Answer: a) Charge remains constant

PRACTICAL PROBLEMS:

31. When multiple dielectrics are used in a capacitor:

- a) Average of dielectric constants is used
- b) Sum of dielectric constants is used

- c) Effective dielectric constant must be calculated
- d) Only highest dielectric constant matters

Answer: c) Effective dielectric constant must be calculated

- 32. In charging a capacitor, the time taken depends on:**
- a) Only capacitance
 - b) Only voltage
 - c) RC time constant
 - d) Only resistance

Answer: c) RC time constant

SYSTEM BEHAVIOR:

- 33. When two charged capacitors of different voltages are connected:**
- a) Charges add
 - b) Voltages add
 - c) Charge redistributes
 - d) Nothing happens

Answer: c) Charge redistributes

- 34. The force between capacitor plates:**
- a) Increases with charge
 - b) Decreases with separation
 - c) Is always attractive
 - d) All of above

Answer: d) All of above

- 35. During discharge of a capacitor:**
- a) Energy is lost as heat

- b) Energy is converted to magnetic field
- c) Energy is conserved perfectly
- d) Energy increases

Answer: a) Energy is lost as heat

APPLICATIONS IN CIRCUITS:

36. Capacitors in AC circuits:

- a) Block DC
- b) Pass AC
- c) Both a and b
- d) Neither a nor b

Answer: c) Both a and b

37. The purpose of capacitor bank in power systems is:

- a) Energy storage
- b) Power factor correction
- c) Voltage regulation
- d) All of above

Answer: d) All of above

SAFETY AND LIMITATIONS:

38. A charged capacitor remains dangerous because:

- a) Current continues to flow
- b) Stored energy remains
- c) Voltage increases
- d) Resistance decreases

Answer: b) Stored energy remains

39. Maximum energy density in a capacitor is limited by:

- a) Plate material
- b) Dielectric strength
- c) Temperature
- d) All of above

Answer: d) All of above

40. In real capacitors, leakage current is due to:

- a) Non-perfect dielectric
- b) Plate oxidation
- c) External fields
- d) Temperature only

Answer: a) Non-perfect dielectric

Lecture 26

BASIC CONCEPTS:

1. Electric current is defined as:

- a) Flow of electrons only
- b) Flow of protons only
- c) Rate of flow of charge
- d) Movement of atoms

Answer: c) Rate of flow of charge

2. The SI unit of current, ampere, is defined as:

- a) One coulomb per second
- b) One volt per second
- c) One joule per coulomb
- d) One coulomb per volt

Answer: a) One coulomb per second

CURRENT DIRECTION:

3. In a conductor, the conventional current flow is:

- a) Same as electron flow**
- b) Opposite to electron flow**
- c) Perpendicular to electron flow**
- d) Independent of electron flow**

Answer: b) Opposite to electron flow

OHM'S LAW:

4. When does Ohm's law strictly apply?

- a) Always**
- b) Only for metals**
- c) When V-I graph is linear**
- d) Only at high voltages**

Answer: c) When V-I graph is linear

5. The resistance R in Ohm's law $V=IR$ is:

- a) Always constant**
- b) Always variable**
- c) Constant only if Ohm's law applies**
- d) Never constant**

Answer: c) Constant only if Ohm's law applies

CIRCUIT ANALYSIS:

6. According to Kirchhoff's voltage law:

- a) Total current is zero**
- b) Sum of voltage drops equals zero**

- c) Resistance is constant
- d) Current is constant

Answer: b) Sum of voltage drops equals zero

POWER AND ENERGY:

7. The power dissipated in a resistor can be expressed as:

- a) Only V^2/R
- b) Only I^2R
- c) Only VI
- d) All of the above

Answer: d) All of the above

8. One watt is equal to:

- a) One joule per second
- b) One volt per ampere
- c) One volt-ampere
- d) All of the above

Answer: d) All of the above

SERIES AND PARALLEL CIRCUITS:

9. For resistors in series:

- a) Currents add
- b) Voltages add
- c) Resistances divide
- d) All voltages are equal

Answer: b) Voltages add

CURRENT DENSITY:

10. Current density J is defined as:

- a) Current per unit volume
- b) Current per unit area
- c) Total current in conductor
- d) Rate of charge flow

Answer: b) Current per unit area

11. The drift velocity of charges in a conductor:

- a) Equals the speed of light
- b) Is typically very high
- c) Is typically very low
- d) Is independent of current

Answer: c) Is typically very low

PRACTICAL APPLICATIONS:

12. When current flows through a resistor:

- a) Its resistance decreases
- b) Its resistance increases due to heating
- c) Its resistance remains constant
- d) Its resistance becomes zero

Answer: b) Its resistance increases due to heating

ADVANCED CONCEPTS:

13. In an RC circuit, the time constant τ is:

- a) $R+C$
- b) $R \times C$
- c) R/C
- d) C/R

Answer: b) $R \times C$

14. The current in an RC circuit:

- a) Increases exponentially**
- b) Decreases exponentially**
- c) Remains constant**
- d) Varies sinusoidally**

Answer: b) Decreases exponentially

15. Conservation of charge in circuits means:

- a) Total charge is zero**
- b) Current entering equals current leaving a junction**
- c) Voltage must be conserved**
- d) Power must be conserved**

Answer: b) Current entering equals current leaving a junction

ADVANCED CIRCUIT ANALYSIS:

16. In a complex circuit with multiple loops, the correct approach is to:

- a) Apply only Ohm's law**
- b) Use only current division**
- c) Apply Kirchhoff's laws systematically**
- d) Consider only voltage drops**

Answer: c) Apply Kirchhoff's laws systematically

17. The drift velocity in a conductor is affected by:

- a) Only current**
- b) Only cross-sectional area**
- c) Only charge carrier density**
- d) All of the above**

Answer: d) All of the above

PRACTICAL MEASUREMENTS:

18. When measuring current in a circuit:

- a) Ammeter is connected in series**
- b) Ammeter is connected in parallel**
- c) Position doesn't matter**
- d) Multiple ammeters are needed**

Answer: a) Ammeter is connected in series

19. For accurate voltage measurement:

- a) Voltmeter should have infinite resistance**
- b) Voltmeter should have zero resistance**
- c) Resistance doesn't matter**
- d) Voltage is independent of measurement**

Answer: a) Voltmeter should have infinite resistance

20. The relationship between power and current is non-linear because:

- a) $P = I^2R$**
- b) Resistance varies with temperature**
- c) Both a and b**
- d) Neither a nor b**

Answer: c) Both a and b

THEORETICAL CONCEPTS:

21. In a multi-loop circuit, the number of independent equations needed equals:

- a) Number of resistors**

- b) Number of voltage sources
- c) Number of independent loops
- d) Number of junctions

Answer: c) Number of independent loops

22. The exponential decay in RC circuits occurs because:

- a) Resistance decreases
- b) Capacitor charge affects voltage
- c) Current follows Ohm's law
- d) Temperature changes

Answer: b) Capacitor charge affects voltage

PRACTICAL APPLICATIONS:

23. In household wiring, parallel connections are used because:

- a) It's cheaper
- b) Devices can operate independently
- c) Current is lower
- d) Voltage is higher

Answer: b) Devices can operate independently

24. The effect of temperature on resistance is important because:

- a) It affects current flow
- b) It can cause device failure
- c) It changes power consumption
- d) All of the above

Answer: d) All of the above

25. Current density is highest:

- a) In thick wires

- b) In thin wires
- c) At wire ends
- d) Near voltage source

Answer: b) In thin wires

SAFETY AND LIMITATIONS:

26. Maximum current in a wire is limited by:

- a) Voltage only
- b) Resistance only
- c) Heat dissipation
- d) Wire length only

Answer: c) Heat dissipation

27. In real circuits, power is lost mainly through:

- a) Magnetic effects
- b) Heat generation
- c) Light emission
- d) Sound production

Answer: b) Heat generation

MATHEMATICAL RELATIONSHIPS:

28. If current doubles, power dissipation:

- a) Doubles
- b) Quadruples
- c) Halves
- d) Remains same

Answer: b) Quadruples

29. The time constant in an RC circuit represents:

- a) Total discharge time**
- b) Time for current to reach zero**
- c) Time for current to fall to $1/e$ of initial value**
- d) Charging time**

Answer: c) Time for current to fall to $1/e$ of initial value

30. For a given voltage, increasing resistance:

- a) Increases current**
- b) Decreases current**
- c) Doesn't affect current**
- d) Makes current oscillate**

Answer: b) Decreases current

MICROSCOPIC ASPECTS:

31. At microscopic level, resistance occurs due to:

- a) Electron collisions with atoms**
- b) Voltage variations**
- c) Current fluctuations**
- d) Temperature only**

Answer: a) Electron collisions with atoms

32. The drift velocity of electrons is:

- a) Same as current velocity**
- b) Much less than electron thermal velocity**
- c) Equal to speed of light**
- d) Independent of applied voltage**

Answer: b) Much less than electron thermal velocity

SYSTEM BEHAVIOR:

33. In steady state:

- a) Current must be constant
- b) Voltage must be constant
- c) Both current and voltage must be constant
- d) Neither needs to be constant

Answer: c) Both current and voltage must be constant

34. Junction rule in circuits follows from:

- a) Ohm's law
- b) Conservation of charge
- c) Conservation of energy
- d) Kirchhoff's voltage law

Answer: b) Conservation of charge

35. The potential difference across parallel branches:

- a) Adds up
- b) Is the same
- c) Depends on current
- d) Must be zero

Answer: b) Is the same

Lecture 27

BASIC CONCEPTS:

1. The magnetic force on a moving charged particle depends on:

- a) Only charge
- b) Only velocity

- c) Only magnetic field strength
- d) All of the above

Answer: d) All of the above

2. When velocity and magnetic field are parallel, the magnetic force is:

- a) Maximum
- b) Minimum
- c) Zero
- d) Infinite

Answer: c) Zero

UNITS AND MEASUREMENTS:

3. One Tesla is equivalent to:

- a) 1 N/C
- b) 1 N/(A·m)
- c) 1 N·m
- d) 1 N/A

Answer: b) 1 N/(A·m)

4. The Earth's magnetic field strength is approximately:

- a) 1 T
- b) 10^{-4} T
- c) 1 Gauss
- d) 5 T

Answer: b) 10^{-4} T

LORENTZ FORCE:

5. The Lorentz force is:

- a) Always perpendicular to velocity
- b) Sum of electric and magnetic forces
- c) Always parallel to magnetic field
- d) Independent of charge

Answer: b) Sum of electric and magnetic forces

6. In a velocity selector, particles move undeflected when:

- a) $v = E/B$
- b) $v = EB$
- c) $v = E+B$
- d) $v = E-B$

Answer: a) $v = E/B$

PARTICLE MOTION:

7. A charged particle in a uniform magnetic field moves in:

- a) A straight line
- b) A circular path
- c) A spiral path
- d) Depends on initial velocity

Answer: b) A circular path

8. The radius of circular motion in magnetic field depends on:

- a) Only charge
- b) Only mass
- c) Only velocity
- d) All of above

Answer: d) All of above

ENERGY CONSIDERATIONS:

9. Magnetic force does no work because:

- a) Force is too small**
- b) Force is perpendicular to displacement**
- c) Force is parallel to displacement**
- d) Force is zero**

Answer: b) Force is perpendicular to displacement

10. The speed of a charged particle in uniform magnetic field:

- a) Increases**
- b) Decreases**
- c) Remains constant**
- d) Oscillates**

Answer: c) Remains constant

APPLICATIONS:

11. Mass spectrometer works on the principle of:

- a) Electric deflection**
- b) Magnetic deflection**
- c) Both a and b**
- d) Neither a nor b**

Answer: b) Magnetic deflection

12. Van Allen belts are formed due to:

- a) Earth's gravity**
- b) Earth's magnetic field**
- c) Solar radiation**
- d) Atmospheric pressure**

Answer: b) Earth's magnetic field

CURRENT-CARRYING CONDUCTORS:

13. The force on a current-carrying wire in magnetic field depends on:

- a) Current only**
- b) Field strength only**
- c) Length only**
- d) Product of all three**

Answer: d) Product of all three

14. The magnetic moment of a current loop is:

- a) IA**
- b) I/A**
- c) $I+A$**
- d) $I-A$**

Answer: a) IA

AMPERE'S LAW:

15. According to Ampere's law, line integral of B around a closed loop equals:

- a) Zero**
- b) Total current enclosed**
- c) Net charge enclosed**
- d) Electric field enclosed**

Answer: b) Total current enclosed

ADVANCED CONCEPTS:

16. When a charged particle enters a magnetic field at an angle θ , its path is:

- a) Circular
- b) Helical
- c) Parabolic
- d) Linear

Answer: b) Helical

17. The frequency of circular motion in magnetic field (cyclotron frequency) is:

- a) $qB/2\pi m$
- b) qB/m
- c) mB/q
- d) $2\pi qB/m$

Answer: a) $qB/2\pi m$

FIELD INTERACTIONS:

18. In a region with both E and B fields, a charged particle can move undeflected when:

- a) $E > B$
- b) $E < B$
- c) $E = vB$
- d) $E = B$

Answer: c) $E = vB$

19. The magnetic force on a current-carrying wire is maximum when:

- a) Wire is parallel to field
- b) Wire is perpendicular to field

- c) Wire is at 45° to field
- d) Independent of angle

Answer: b) Wire is perpendicular to field

20. For a charged particle moving in a magnetic field, which quantity remains constant?

- a) Velocity
- b) Speed
- c) Direction
- d) Acceleration

Answer: b) Speed

PRACTICAL APPLICATIONS:

21. The principle of mass spectrometer is based on:

- a) Different charges
- b) Different velocities
- c) Different masses
- d) Different accelerations

Answer: c) Different masses

22. In a cyclotron, the magnetic field is used to:

- a) Accelerate particles
- b) Keep particles in circular path
- c) Increase particle energy
- d) Decrease particle energy

Answer: b) Keep particles in circular path

FIELD BEHAVIOR:

23. Outside a long straight wire carrying current I , the magnetic field B varies as:

- a) $1/r$
- b) $1/r^2$
- c) r
- d) r^2

Answer: a) $1/r$

24. Inside a current-carrying wire, the magnetic field:

- a) Is zero
- b) Increases linearly with r
- c) Decreases with r
- d) Is constant

Answer: b) Increases linearly with r

25. The magnetic field lines around a straight current-carrying conductor are:

- a) Parallel to conductor
- b) Perpendicular to conductor
- c) Concentric circles
- d) Radial lines

Answer: c) Concentric circles

THEORETICAL ANALYSIS:

26. The magnetic force on a charged particle can:

- a) Change its energy
- b) Change its direction
- c) Change both energy and direction
- d) Change neither energy nor direction

Answer: b) Change its direction

27. For particles with same velocity in a magnetic field, the radius of circular path depends on:

- a) q/m ratio
- b) mq ratio
- c) m/q ratio
- d) q^2/m ratio

Answer: c) m/q ratio

EXPERIMENTAL CONSIDERATIONS:

28. In magnetic field measurements, Tesla is equivalent to:

- a) Weber/meter²
- b) Newton/ampere-meter
- c) Both a and b
- d) Neither a nor b

Answer: c) Both a and b

29. The Earth's magnetic field is important for:

- a) Navigation only
- b) Protection from solar wind only
- c) Both a and b
- d) Neither a nor b

Answer: c) Both a and b

30. The direction of magnetic force is determined by:

- a) Left-hand rule
- b) Right-hand rule
- c) Fleming's rule
- d) Coulomb's law

Answer: b) Right-hand rule

ADVANCED APPLICATIONS:

31. The principle of magnetic confinement in fusion reactors uses:

- a) Electric forces only**
- b) Magnetic forces only**
- c) Both electric and magnetic forces**
- d) Neither electric nor magnetic forces**

Answer: b) Magnetic forces only

32. In a Hall effect sensor, the measured voltage is due to:

- a) Electric field only**
- b) Magnetic field only**
- c) Combined effect of E and B fields**
- d) Neither E nor B fields**

Answer: c) Combined effect of E and B fields

33. The magnetic moment of a current loop determines:

- a) Its orientation in magnetic field**
- b) Its energy in magnetic field**
- c) Both a and b**
- d) Neither a nor b**

Answer: c) Both a and b

34. Ampere's law is most useful for calculating magnetic fields with:

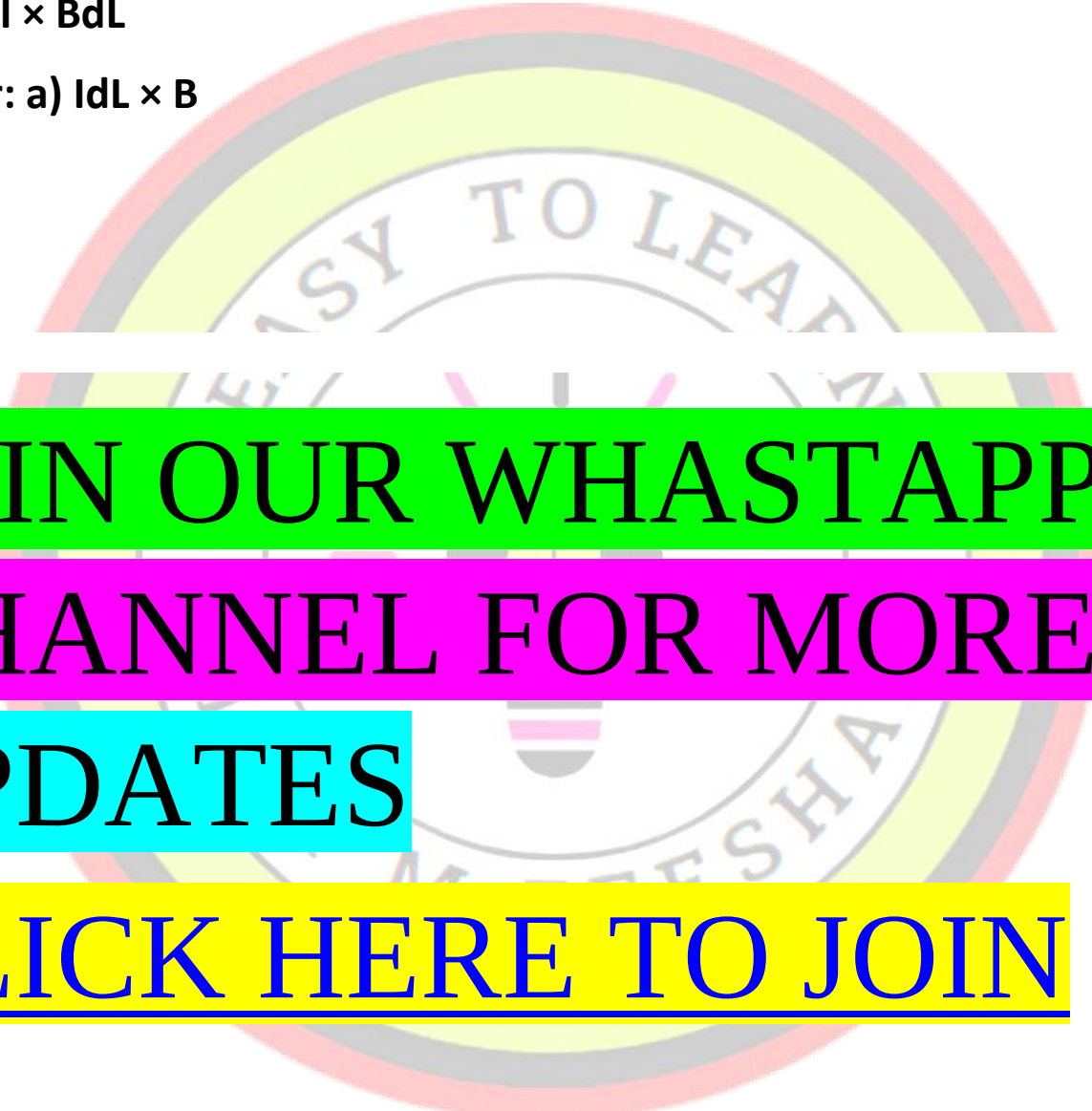
- a) No symmetry**
- b) Spherical symmetry**
- c) Cylindrical symmetry**
- d) Any symmetry**

Answer: c) Cylindrical symmetry

35. The magnetic force on a current element is:

- a) $I d\mathbf{L} \times \mathbf{B}$
- b) $B d\mathbf{L} \times \mathbf{I}$
- c) $\mathbf{B} \times I d\mathbf{L}$
- d) $\mathbf{I} \times B d\mathbf{L}$

Answer: a) $I d\mathbf{L} \times \mathbf{B}$



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